NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

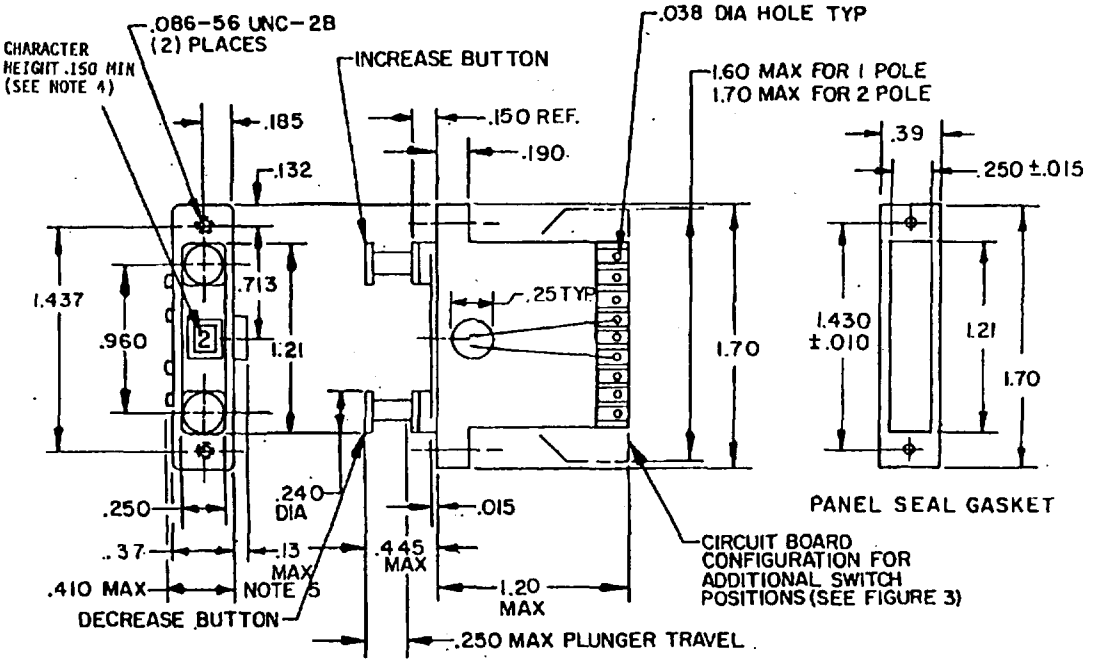
MIL-PRF-22710/18F
30 September 1986
SUPERSEDING
MIL-S-22710/18E
21 May 1981

PERFORMANCE SPECIFICATION SHEET

SWITCHES, CODE, INDICATING WHEEL (PRINTED CIRCUIT) (PUSHBUTTON, BIDIRECTIONAL, SEALED OR UNSEALED, EMI/RFI SHIELDED, ILLUMINATED AND NONILLUMINATED) STYLE SRPC18-1 AND -2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22710.



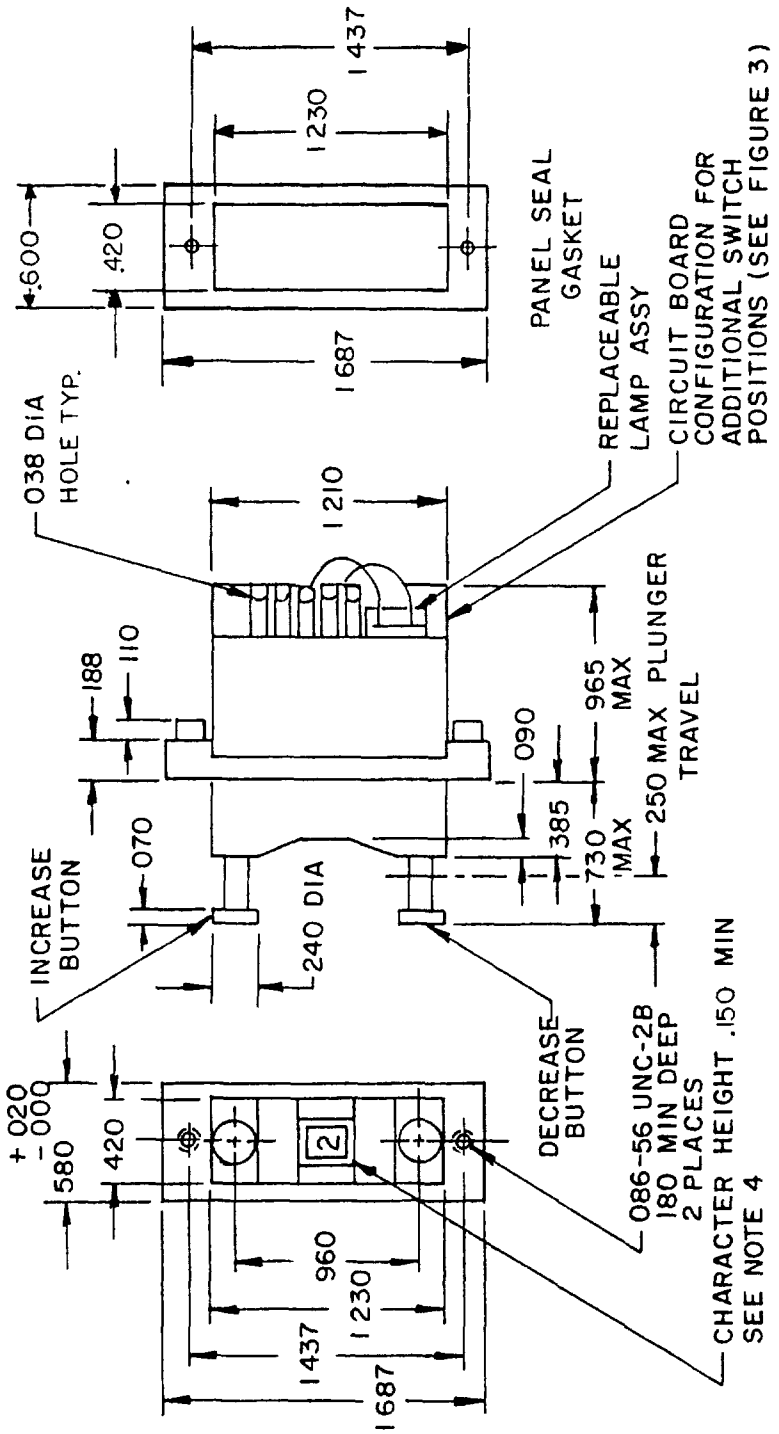
Inches	mm	Inches	mm	Inches	mm
.010	0.25	.190	4.83	.960	24.38
.015	0.38	.240	6.10	1.00	25.4
.031	0.79	.250	6.35	1.02	25.9
.038	0.97	.25	6.4	1.20	30.5
.086	2.18	.370	9.40	1.21	30.7
.13	3.3	.39	9.91	1.43	36.32
.132	3.35	.410	10.41	1.60	40.6
.150	3.81	.445	11.30	1.70	43.2
.185	4.70	.713	18.11		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.02 (.5 mm) for two place decimals.
4. Readout characters: See figure 3 and table I.
5. .13 inch (3.3 mm) maximum dimension applies to light installation only.
6. When a sealed switch configuration is specified, a gasket of silicone rubber .031 inch (.79 mm) thick shall be furnished.

(F) FIGURE 1. Style SRPC18-1 switch.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.



INCHES	MM	INCHES	MM	INCHES	MM
.020	.51	.180	4.57	.730	18 54
.032	.81	.188	4.78	.960	24 38
.038	97	.240	6.10	.965	24 51
.070	1.78	.250	6.35	1.210	30 73
.086	2 18	.385	9.78	1.230	31 24
.090	2.29	420	10.67	1.437	36.50
.110	2 79	.580	14.73	1.687	42.58
.150	3 81	600	15 24		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.02 (.5 mm) for two place decimals.
4. Readout characters. See figure 3 and table I.
5. When a sealed switch configuration is specified, a gasket of silicone rubber .031 inch (.79 mm) thick shall be furnished.

FIGURE 2. Style SRPC[illegible]-2 switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Construction: D.

Angle of throw indexing: 30°, 36°, and 45° indexing.

Temperature-life characteristic: D, test loads as shown in table I. (For lighted switches, maximum temperature is 85°C).

TABLE I. Electrical test load.

Test or condition	Resistive	
At atmospheric pressure	.125 A	28 V dc and 115 V ac
At reduced barometric pressure	0.05 A	28 V dc and 115 V ac
Dry circuit	10 mA	50 mV

Vibration grade: 3 (15g, 10-2,000 Hz).

Shock type: M (100 g's).

Altitude rating: C (70,000 feet).

Terminal design: See figure 3.

Contact resistance:

Initial: 50 milliohms, maximum.

Following life (rotational): 70 milliohms, maximum.

Rotational torque (push force required to index from any detent position to the next): 20 ounces, minimum to 40 ounces, maximum.

Seal: Method II, when specified in part no. code.

Dielectric withstanding voltage:

At atmospheric pressure: 1,000 volts, rms.

At reduced barometric pressure: 450 volts, rms.

Application of test voltage: Between the switch terminals and terminals to case.

Sand and dust: Applicable.

Explosion: Applicable.

Life (rotational): Test-condition letter G, method 206 of MIL-STD-202. Lamps shall be energized at their maximum rated voltage.

Ⓕ EMI/RFI shielding: Applicable when specified (see table III).

Military part number: See part number code.

Luminance: Illuminated and nonilluminated as shown in table II.

TABLE II. Luminance.

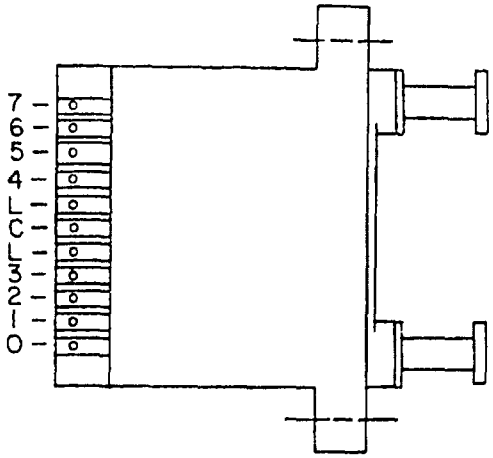
Lamp	Switch filter	Brightness	Rating	
			Volts	Amperes
MS90452-6833	Clear	1.50 ±.75 foot-lamberts	5	.060
MS90452-6833	Red	.75 ±.50 foot-lamberts	5	.060

Lamp failure: Lamp failure shall not be considered a unit failure, if when the lamp failure is noted immediately a new lamp is installed and the lamp functions. Test shall then be continued, as required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

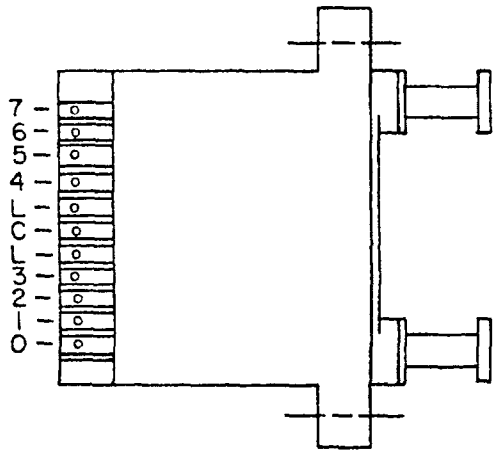
NO.1 45° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS							
	0	1	2	3	4	5	6	7
0	●							
1		●						
2			●					
3				●				
4					●			
5						●		
6							●	
7								●



NO 2 45° INDEXING
CONTACTS ARE SHORTING
MAKE BEFORE BREAK

DIAL READS	COMMON C CONNECTED TO TERMINALS							
	0	1	2	3	4	5	6	7
0	●							
1		●						
2			●					
3				●				
4					●			
5						●		
6							●	
7								●



NO 3 45° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS		
	1	2	4
0			
1	●		
2		●	
3	●	●	
4			●
5	●		●
6		●	●
7	●	●	●

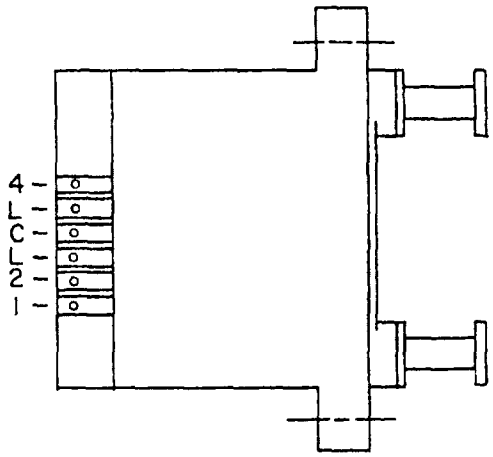
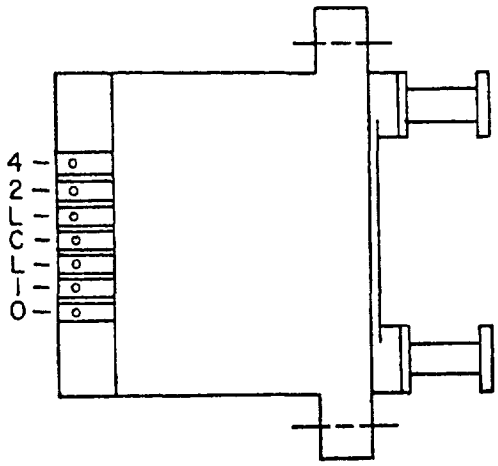


FIGURE 3. Terminal design and configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

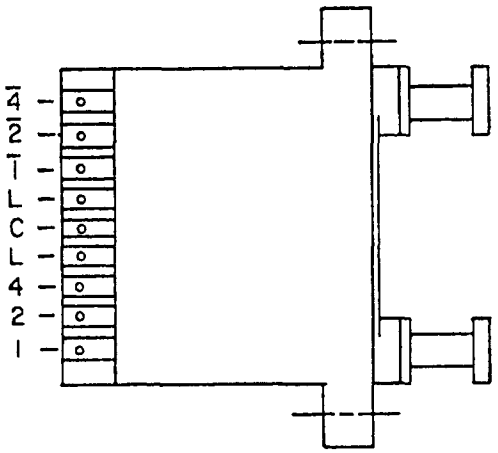
NO.4 45° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS			
	0	1	2	4
0	●			
1		●		
2			●	
3		●	●	
4				●
5		●		●
6			●	●
7		●	●	●



NO.5 45° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS					
	1	2	4	$\bar{1}$	$\bar{2}$	$\bar{4}$
0				●	●	●
1	●				●	●
2		●		●		●
3	●	●				●
4			●	●	●	
5	●		●		●	
6		●	●	●		
7	●	●	●			



NO.6 36° INDEXING

DIAL READS	COMMON C CONNECTION TO TERMINAL									
	0	1	2	3	4	5	6	7	8	9
0	●									
1		●								
2			●							
3				●						
4					●					
5						●				
6							●			
7								●		
8									●	
9										●

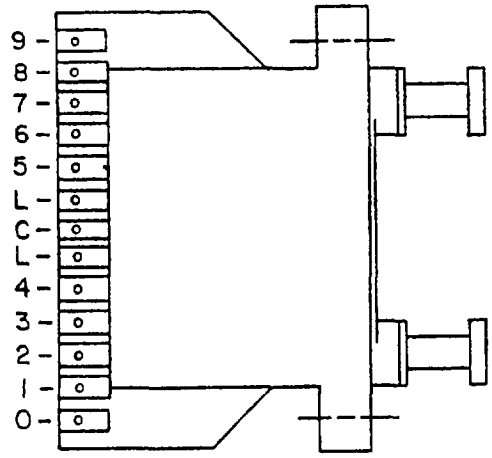
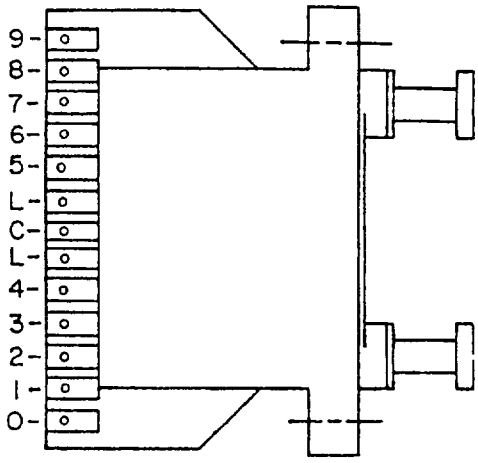


FIGURE 3. Terminal design and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

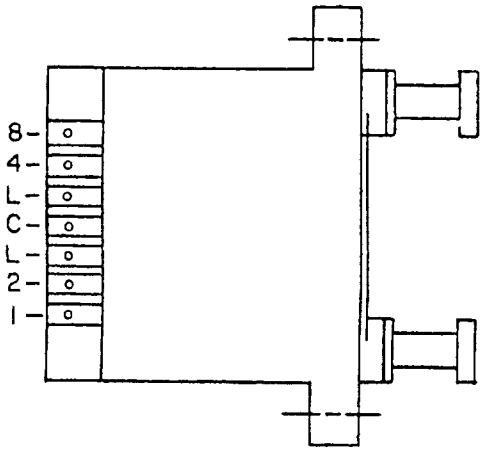
NO 7 36° INDEXING
CONTACTS ARE SHORTING
MAKE BEFORE BREAK

DIAL READS	COMMON C CONNECTION TO TERMINAL									
	0	1	2	3	4	5	6	7	8	9
0	●									
1		●								
2			●							
3				●						
4					●					
5						●				
6							●			
7								●		
8									●	
9										●



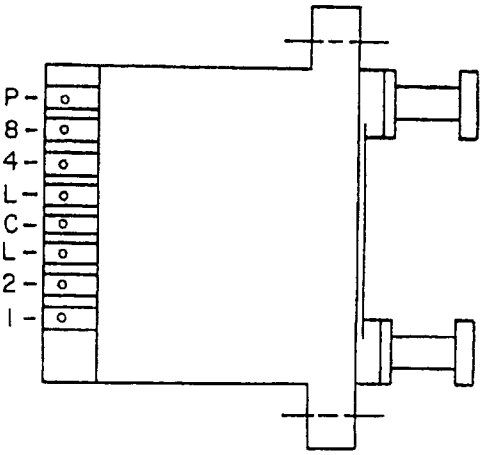
NO 8 36° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●



NO 9 36° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS				
	1	2	4	8	P
0					●
1	●				
2		●			
3	●	●			●
4			●		
5	●		●		●
6		●	●		●
7	●	●	●		
8				●	
9	●			●	●

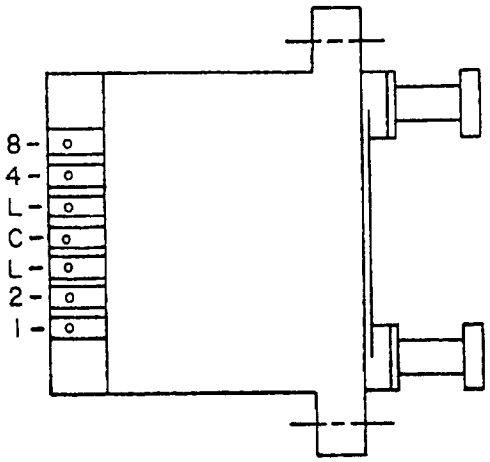


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

FIGURE 3. Terminal design and configuration - Continued.

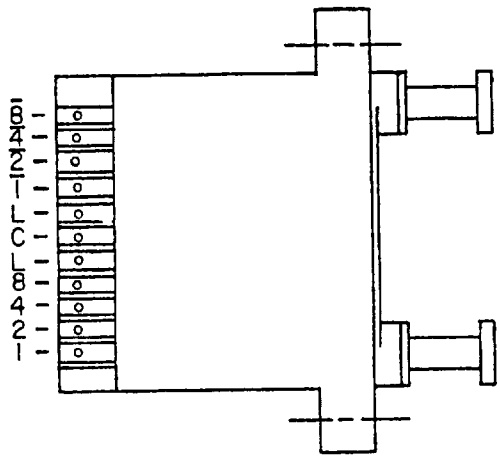
NO.10 36° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS			
	1	2	4	8
0	●	●		
1			●	
2	●		●	
3		●	●	
4	●	●	●	
5				●
6	●			●
7		●		●
8	●	●		●
9			●	●



NO.11 36° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS							
	1	2	4	8	$\overline{1}$	$\overline{2}$	$\overline{4}$	$\overline{8}$
0					●	●	●	●
1	●					●	●	●
2		●			●		●	●
3	●	●					●	●
4			●		●	●		●
5	●		●			●		●
6		●	●		●			●
7	●	●	●					●
8				●	●	●	●	
9	●			●		●	●	



NO 12 30° INDEXING

DIAL READS	COMMON C CONNECTOR TO TERMINAL											
	0	1	2	3	4	5	6	7	8	9	10	11
0	●											
1		●										
2			●									
3				●								
4					●							
5						●						
6							●					
7								●				
8									●			
9										●		
10											●	
11												●

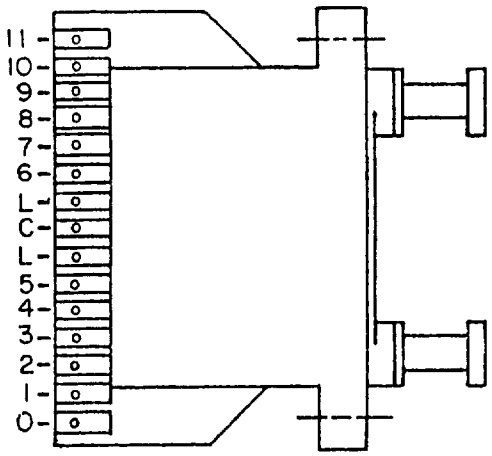
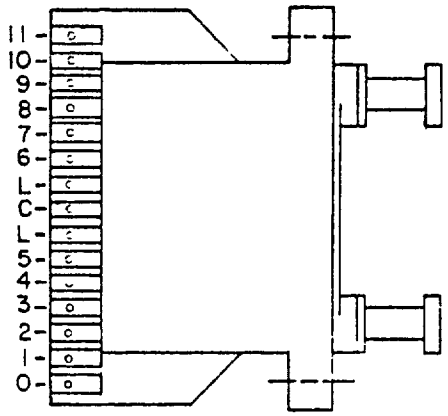


FIGURE 3. Terminal design and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

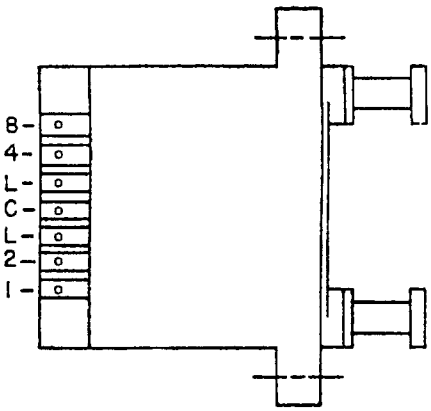
NO 13 30° INDEXING
CONTACTS ARE SHORTING
MAKE BEFORE BREAK

DIAL READS	COMMON C CONNECTOR TO TERMINAL											
	0	1	2	3	4	5	6	7	8	9	10	11
0	●											
1		●										
2			●									
3				●								
4					●							
5						●						
6							●					
7								●				
8									●			
9										●		
10											●	
11												●



NO 14 30° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●
10		●		●
11	●	●		●



NO 15 30° INDEXING

DIAL READS	COMMON C CONNECTED TO TERMINALS							
	1	2	4	8	$\bar{1}$	$\bar{2}$	$\bar{4}$	$\bar{8}$
0					●	○	○	●
1	●					○	○	○
2		●			○		○	○
3	●	●					○	○
4			●		●	●		●
5	●		●			○		○
6		●	●		○			●
7	●	●	●					○
8				●	●	○	○	
9	●			●		●	○	
10		●		●	●		○	
11	●	●		●			●	

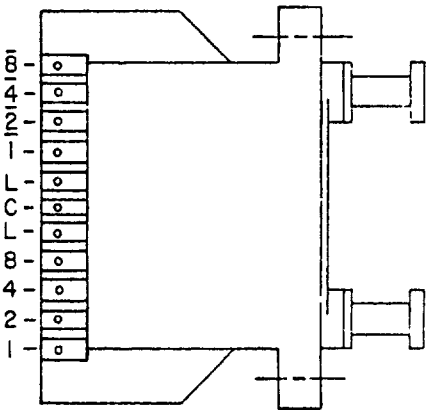
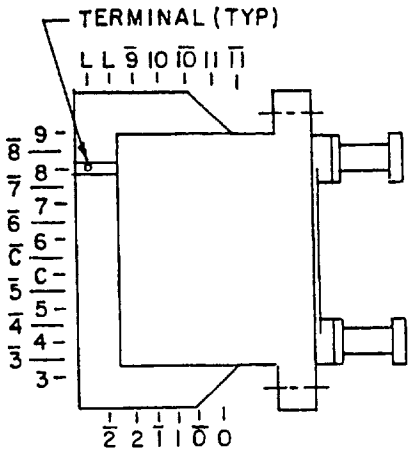


FIGURE 3. Terminal design and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

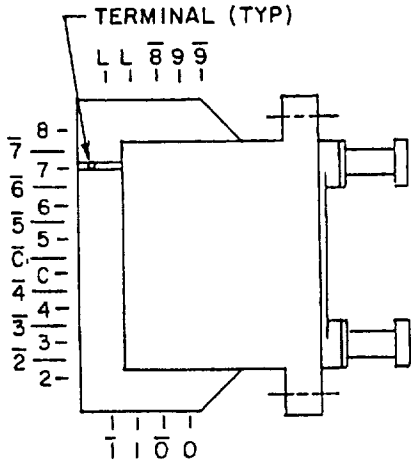
NO. 16 30° INDEXING
2 POLE

DIAL READS	COMMON C CONNECTED TO TERMINAL												DIAL READS	COMMON $\bar{C}$ CONNECTED TO TERMINAL											
	0	1	2	3	4	5	6	7	8	9	10	11		$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{6}$	$\bar{7}$	$\bar{8}$	$\bar{9}$	$\bar{10}$	$\bar{11}$
0	●												0	●											
1		●											1		●										
2			●										2			●									
3				●									3				●								
4					●								4					●							
5						●							5						●						
6							●						6							●					
7								●					7								●				
8									●				8									●			
9										●			9										●		
10											●		10											●	
11												●	11												●



NO 17 30° INDEXING
2 POLE

DIAL READS	COMMON C CONNECTED TO TERMINAL										DIAL READS	COMMON $\bar{C}$ CONNECTED TO TERMINAL									
	0	1	2	3	4	5	6	7	8	9		$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{6}$	$\bar{7}$	$\bar{8}$	$\bar{9}$
0	●										0	●									
1		●									1		●								
2			●								2			●							
3				●							3				●						
4					●						4					●					
5						●					5						●				
6							●				6							●			
7								●			7								●		
8									●		8									●	
9										●	9										●



NO. 18 45° INDEXING
2 POLE

DIAL READS	COMMON C CONNECTED TO TERMINAL								DIAL READS	COMMON $\bar{C}$ CONNECTED TO TERMINAL							
	0	1	2	3	4	5	6	7		$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{6}$	$\bar{7}$
0	●								0	●							
1		●							1		●						
2			●						2			●					
3				●					3				●				
4					●				4					●			
5						●			5						●		
6							●		6							●	
7								●	7								●

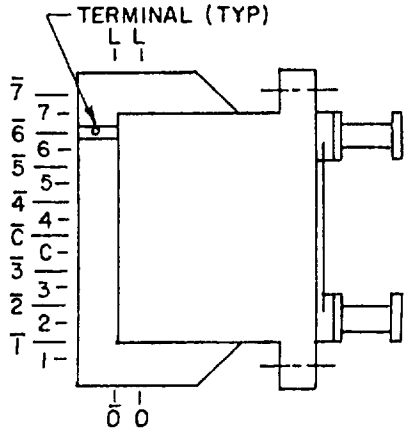


FIGURE 3. Terminal design and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

MILITARY PART NUMBER CODE

M22710/18-1 S 12 A

"1" Style SRPC/18-1 (when illuminated uses one lamp)

"2" Style SRPC/18-2 (when illuminated uses two lamps)

"S" Indicates sealed method II panel seal

"U" Indicates unsealed

Terminal design and construction 1 - 18

Color of stamping of wheel and light requirement table III

Ⓕ TABLE III. Color of stamping and EMI/RFI shielding requirements.

Code	Readout characters	Type of lamp	EMI/RFI shielded
A	1/ B/W 2/	None	No
B	W/B	None	No
C	B/W	C 3/	No
D	W/B	C	No
E	B/W	R 4/	No
F	W/B	R	No
G	B/W	None	Yes
H	W/B	None	Yes
J	B/W	C	Yes
K	W/B	C	Yes
L	B/W	R	Yes
M	W/B	R	Yes

1/ Color of stamping on wheel B (black) or W (white)
2/ Color of background B (black) or W (white)
3/ Color of lamp C (clear)
4/ Color of lamp R (red)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.

TABLE IV. Supersession data.

Superseded military part number M22710/18-	Superseding military part number M22710/18-	Superseded military part number M22710/18-	Superseding military part number M22710/18-	Superseded military part number M22710/18-	Superseding military part number M22710/18-
001	1U1A	025	1U13A	049	1S10A
002	1U1B	026	1U13B	050	1S10B
003	1U2A	027	1U14A	051	1S11A
004	1U2B	028	1U14B	052	1S11B
005	1U3A	029	1U15A	053	1S12A
006	1U3B	030	1U15B	054	1S12B
007	1U4A	031	1S1A	055	1S13A
008	1U4B	032	1S1B	056	1S13B
009	1U5A	033	1S2A	057	1S14A
010	1U5B	034	1S2B	058	1S14B
011	1U6A	035	1S3A	059	1S15A
012	1U6B	036	1S3B	060	1S15B
013	1U7A	037	1S4A	061	1U16A
014	1U7B	038	1S4B	062	1S16A
015	1U8A	039	1S5A	063	1U16B
016	1U8B	040	1S5B	064	1S16B
017	1U9A	041	1S6A	065	1U17A
018	1U9B	042	1S6B	066	1S17A
019	1U10A	043	1S7A	067	1U17B
020	1U10B	044	1S7B	068	1S17B
021	1U11A	045	1S8A	069	1U18A
022	1U11B	046	1S8B	070	1S18A
023	1U12A	047	1S9A	071	1U18B
024	1U12B	048	1S9B	072	1S18B

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - SM
Navy -
Air Force - 99
DLA - ES

User activities:
Army - ME, AL
Navy - MC, AS
Air Force - 11, 17

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-1309-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:33Z
Check the source to verify that this is the current version before use.